Predicting Rio Grande Drying with RiverEyes Data

Stakeholder Driven Data Analysis to Improve Water Sustainability in NM
Technical Report – 11 May 2021

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Plain Language Summary
Low river flows in the Rio Grande result in drying between Isleta diversion dam downstream to Elephant Butte. Developing dry spot predictions can provide managers tools to efficiently deliver water to users while minimizing river drying. To make those prediction, we began identifying factors correlated with the occurrence, persistence, and extent of river drying using river gages, diversion amounts, and estimates of evaporation from plants and open water. Initial findings are based on a case study of the persistence of drying at river mile 74. The Otowi index supply, which is the recorded flow of the Rio Grande at Otowi Bridge, adjusted for storage in reservoirs and trans-mountain diversions predicted the persistence of drying at river mile 74. We predicted that for every 15 kaf increase in the Otowi index supply, drying at river mile 74 decreased by a day. Annual diversions at Isleta dam also predicted the persistence of drying at river mile 74, with one less day of drying predicted to occur for every 1,000 cfs diverted. Identifying whether these relationships are due to direct flow past diversions, because of agriculture returns, or more complicated interactions of system components would be beneficial. Since it is intuitive that higher amounts of water measured as the Otowi index supply results in more water in the river, we recommend it as a better predictor of drying than diversions at the Isleta dam.

Main Text
Introduction
Flows in Rio Grande between Isleta diversion dam and Elephant Butte reservoir can be insufficient to sustain surface water. Reducing the occurrence, persistence, and extent of dry river reaches may appease those who consider a flowing river to hold cultural significance. In addition, dry stretches of river result in reduced ecosystem capacity for sustenance of plant and animals and thus minimizing river drying could result in reduce regulatory burdens. Because the New Mexico Interstate Stream Commission's (NMISC) mission includes the power to "investigate, protect, conserve, and develop New Mexico's waters (NM Stat § 72-14-3)" NMISC collaborates with other agencies to measure and monitor hydrologic factors that affect management of water in the Rio Grande between Isleta diversion dam and Elephant Butte reservoir. This includes measuring diversion amounts, river flows, evaporative loss, precipitation, and other system components. One monitoring effort is the daily identification of river drying when discharge reaches key thresholds (USGS gages at the Rio Grande Floodway at San Acacia is below 300 cubic feet per second or the Rio Grande Near Bosque Farms is below 80) (McKenna 2019). This monitoring effort is called RiverEyes and has systematically mapped the occurrence, persistence, and extent of river drying between Isleta diversion dam and Elephant Butte reservoir since 2002. Correlating hydrologic factors to the RiverEyes data has the potential to provide managers more precises and efficient ways to reduce river drying (Srinivasan et al. 2015).

Hypotheses, predictions, and data sources

For this investigation, I developed four hypothesis and associated predictions to correlate river drying between Isleta diversion dam and Elephant Butte reservoir, as measured by RiverEyes, and some of the system's hydrologic factors (Table1).

Table 1. Hypothesis, predications, and data sources to forecast riverbed drying in the Rio Grande between Isleta diversion dam and Elephant Butte reservoir, New Mexico. Abbreviations are: Middle Rio Grande Conservancy District (MRGCD), U.S. Geological Survey (USGS), U.S. Bureau of Reclamation (BOR), and New Mexico Interstate Stream Commission (NMISC). Diversion dam channels are Belen highline canal (BELCN), Peralta main canal (PERCN), Chical lateral (CHICN), Old Caique Acequia (CHACN), Upper Chical Acequia (CACCN), and San Acacia (SNAN5)

Hypothesis Dry points occur, persist, and extent primarily due to…	**Prediction** Dry point occurrence, length of time dry, and extent of dryness will…	**Predictor and data source**	**Result for drying persistence at river mile 74**
H1: diversions for agriculture removing flows from the main channel	P1: be positively correlated with diversion discharge	MRGCD gages - Isleta dam (BELCN, PERCN, CHICN, CHACN, CACCN)	negatively correlated
		- San Acacia dam (SNAN5)	not correlated
	P2: not be correlated with river discharge	USGS gage - San Acacia	not assessed
	P3: be correlated with agricultural evapotranspiration	BOR ET Toolbox - Agriculture	not assessed
H2: summer increases in evaporation in the natural portion of the system reducing contributions to streamflow	P1: positively correlated with a riparian evapotranspiration	BOR ET Toolbox - Riparian	not assessed
	P2: positively correlated with open water evaporation	BOR ET Toolbox - Open water	not assessed
H3: the amount of water available in the system	P1: be positively correlated with the Otowi index supply	NMISC - Otowi supply index	negatively correlated
H4: low monsoon-derived streamflow because monsoon-derived streamflow drives Rio Grande streamflow at the time of year that dry points most often occur	P1: be negatively correlated with amount of precipitation in June through September.	BOR ET Toolbox - Rainfall	not assessed

Analysis methods

I developed three response variables using 2003-2018 RiverEyes observations These were:

1) daily identification of an individual river mile as either wet or dry (occurrence),
2) annual number of days an individual river mile was dry (persistence), and
3) daily number of dry miles (extent).

The factors I collected to predict river drying were daily evapotranspiration estimates from the U.S. Bureau of Reclamations evapotranspiration (ET) Toolbox, daily U.S. Geological Survey (USGS) stream gages, daily Middle Rio Grande Conservancy District diversion gages, and the annual Otowi index supply. ET Toolbox estimates are calculated for specific reaches of river and required me to bin the RiverEyes data to the appropriate ET Toolbox river reach - reach 5 is river miles 168-129, reach 6 is 128-116, reach 7 is 115-68, and reach 8 is 67-54.

Data visualization identified methodological changes in ET Toolbox modeling (Brower 2018) resulted in a separation of the time series to two periods – 2003 to 2011 and 2012 to 2018 (Figure AI). Applying a mean baseline correction was unsuccessful (Appendix I and Figure A2). Although the ET Toolbox data could be split into two time periods, I removed these from the analysis until an accurate correction of the entire data set can be made. This left three hypotheses/predictions and their associated datasets to test (H1P1, H1P2, and H3P1).

To develop my methodology and as a case study, I queried the data for river mile 74 because it was frequently dry and assessed whether the persistence of drying at this river mile was predictable. Prior to model construction, I summed the total annual diversions at the Isleta diversion dam, San Acacia diversion dam, and river flow at the USGS Rio Grande Floodway at San Acacia (San Acacia gage). I then assessed correlations between predictor variables and removed those that were highly correlated. The Otowi index supply and San Acacia gage were highly correlated, and I chose to remove the San Acacia gage (Appendix II, Figure A3). I then constructed a set of linear models controlling for the non-independence of data collected in the same year by allowing regression intercepts to vary by year and used the function 'lme' from the 'lme4' package (Laird and Ware 1982, Bates 2015). All predictor variables were scaled at this stage (i.e., the standard deviation was divided out of the data). An information-theoretic approach ranked competing models using Akaike's Information Criterion corrected for small sample sizes (AICc) and Bayesian information criterion (BIC) (Akaike, 1973; Hurvich & Tsai, 1989). I retained models for interpretation whose AICc or BIC difference from the top model was < 2.0 using 'AICctab' and 'BICtab' from the 'bbmle' package. For retained models, I tested the predictive significance of factors by calculating confidence intervals (CI) using "intervals' from the 'nlme' package and an analysis of variance (ANOVA) using 'Anova' from the 'car' package. Once significant predictors were identified, I estimated the effect of those predictors using their unscaled values and recalculating CIs and ANOVA p-values. After this methodology was established, I modeled the persistence of drying for individual river miles 155, 134, and 80 (Appendix III). All analyses were run using the R statistical language (R Core Team, 2015)

Results/discussion

Although some river miles of the Rio Grande between the Isleta diversion dam and Elephant Butte reservoir do not dry (occurrence), others often do (Figure 1). Rewetting of the channel appears to occur in sections that are distinguished by returns to the river. Individual river miles dried up to 60 days in a year (persistence) and at times, the length of a dry reach (extent) exceed 35 miles (Figures 2 and 3).

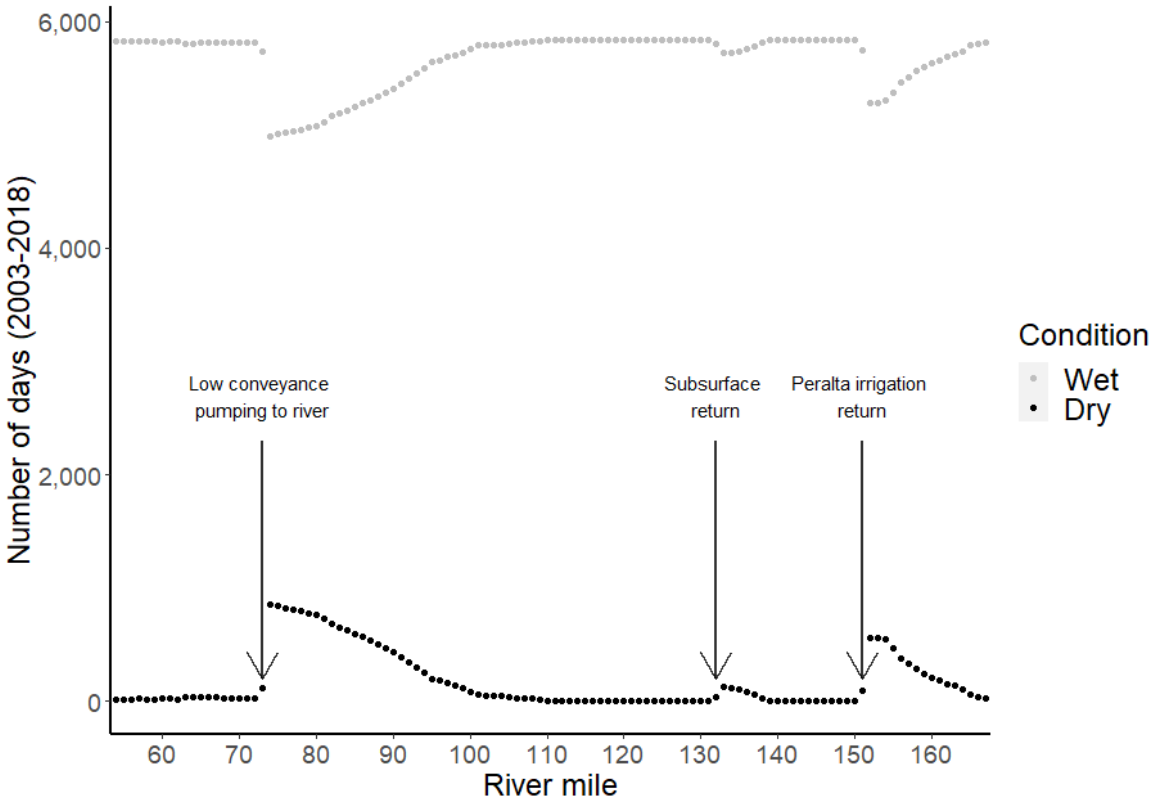

Figure 1. Occurrence and persistence of drying at individual river miles between 2003 and 2018 in the Rio Grande from Isleta diversion dam (mile 168) to Elephant Butte reservoir (mile 54).

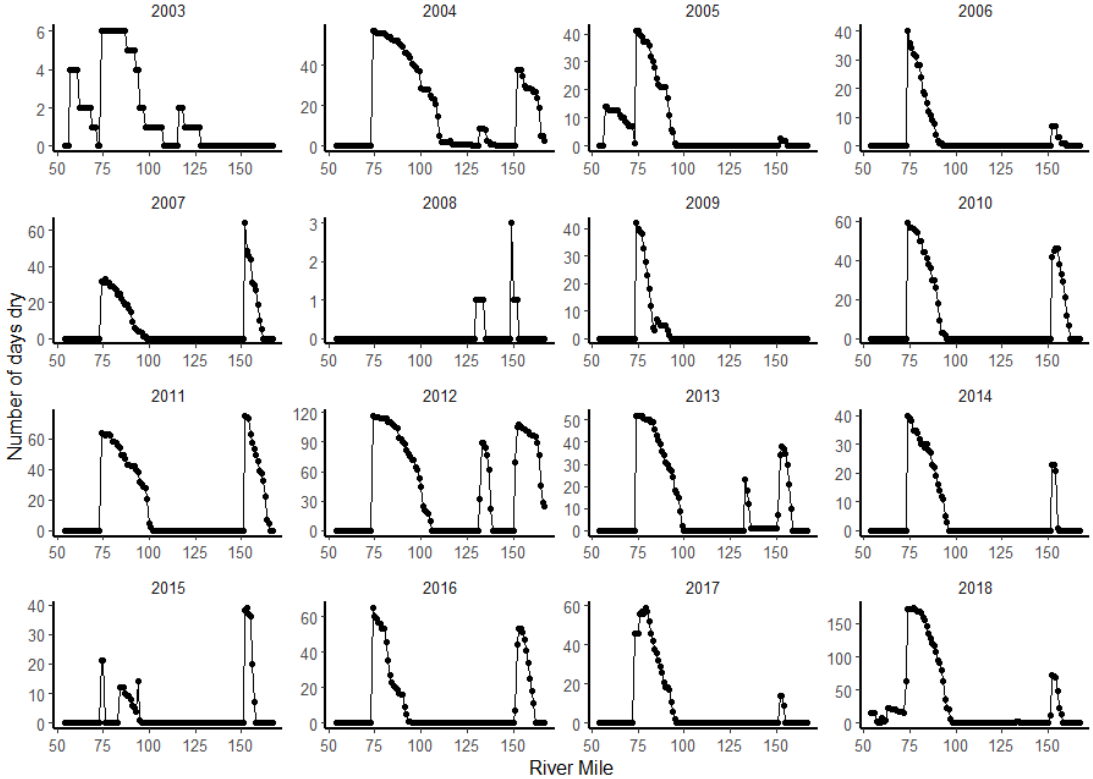

Figure 2. Occurrence and persistence of drying at individual river miles by year in the Rio Grande from Isleta diversion dam (mile 168) to Elephant Butte (mile 54).

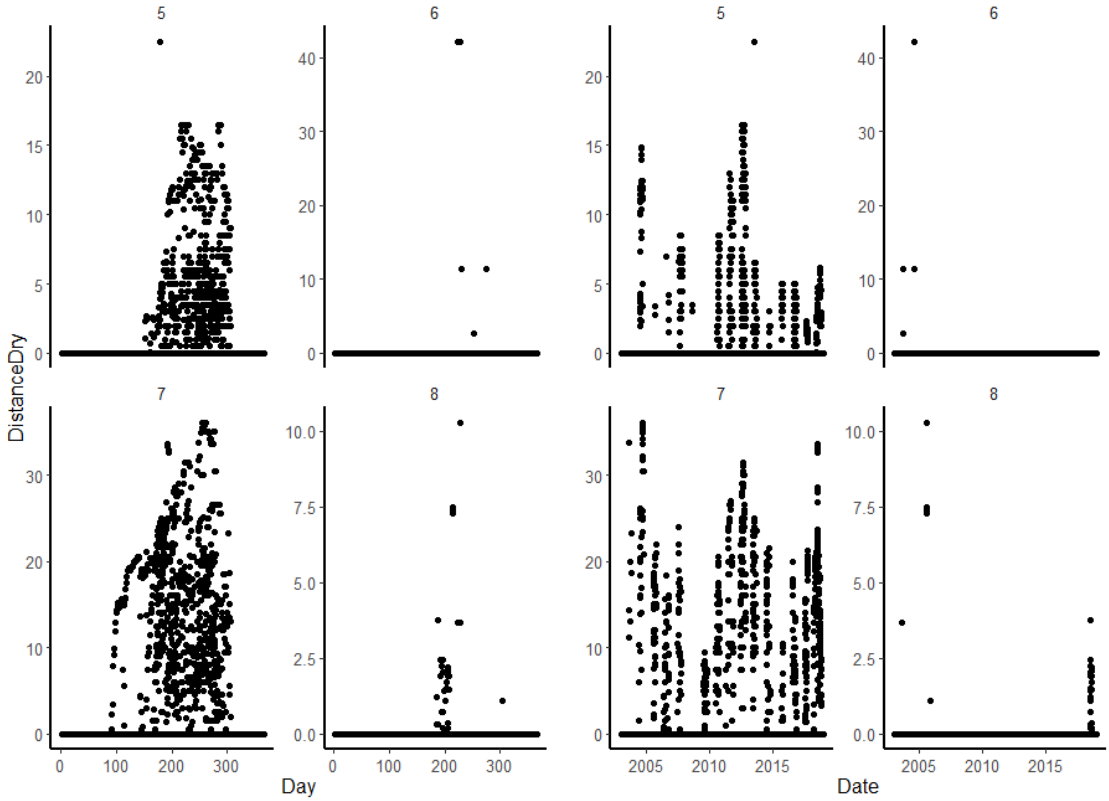

Figure 3. Daily extent of dry river miles between Isleta diversion dam and Elephant Butte reservoir by US Bureau of Reclamation ET Toolbox reaches 5 through 8 by date (left) and day of year (right) from 2003 to 2018.

Our river mile 74 case study suggested that the Otowi index supply and annual diversions at Isleta dam could predict the persistence of drying at this river mile (Appendix II). The relationship with both predictors was negative which indicates that with an increase in the Otowi index supply or annual diversions at Isleta dam there are fewer days of drying (Table 2 and Figure 4). For the Otowi index supply, the model predicted that for every 15 kaf (8 – 250 CI) there is a one-day reduction in drying at river mile 74. For annual diversions at the Isleta dam, every 1,000 cfs (500 - 14,000 CI) predicted one less day of drying. Modeling of river miles 155 and 80 passed statistical assumptions (Appendix III). The top models for these two river miles only included the Otowi index supply and because confidence intervals overlapped zero, the index was not considered a significant predictor of drying at these river miles. The null model was also included in the top model set and this suggests overall modeling fitting was poor. Modeling of river mile 134 did not pass statistical assumptions. These results suggest that including additional hydrologic factors could result in better fitting models and predictions.

Table 2. Estimated effect size and lower and upper confidence intervals (CI) for the top models predicting persistence of drying at Rio Grande mile 74. P-value calculated through an analysis of variance.

Model	Estimated effect	Lower CI	Upper CI	P-value
Otowi index supply (kaf)	-0.067	-0.129	-0.004	0.02
Isleta diversion (cfs)	-0.001	-0.002	-0.00007	0.02

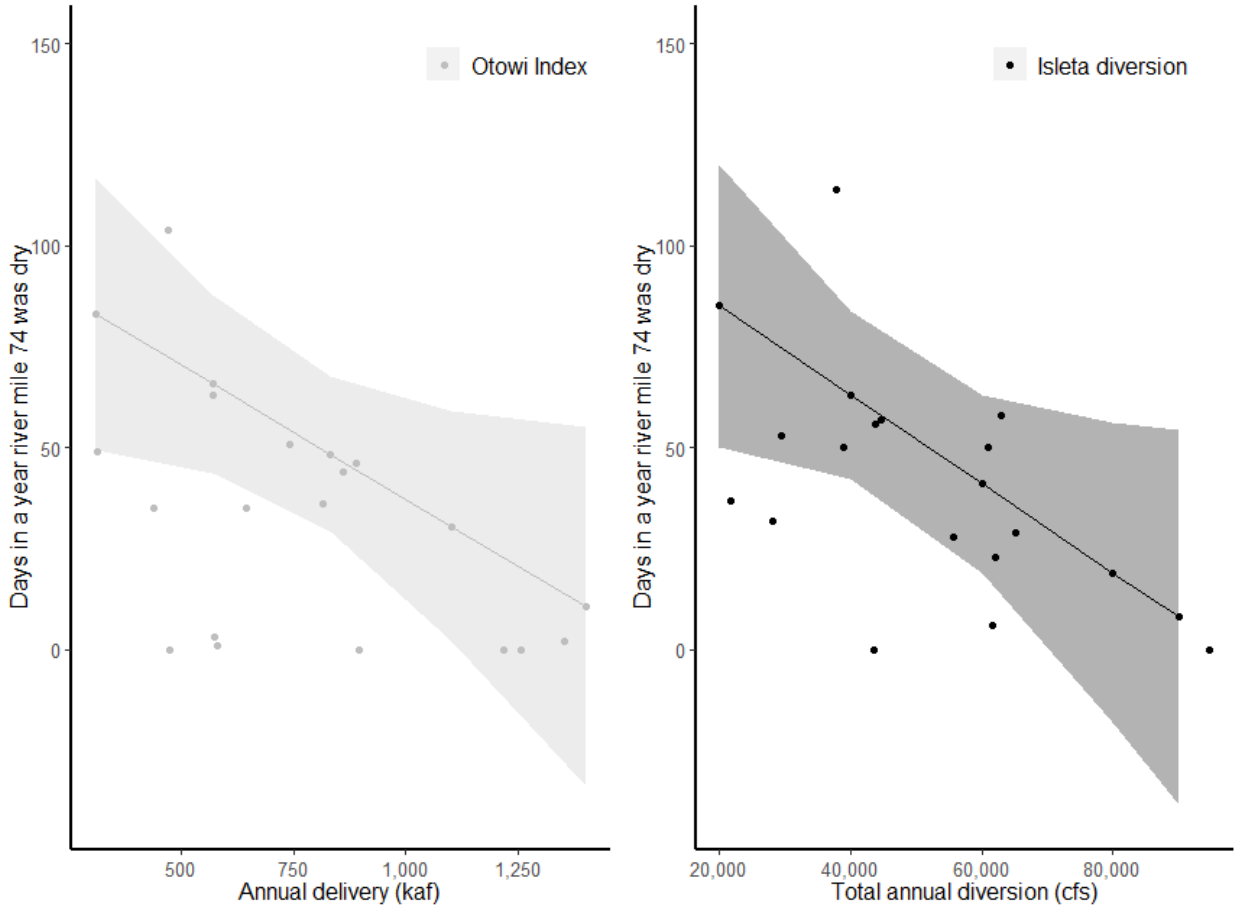

Figure 4. Expectations from the top linear models predicting the number of days Rio Grande mile 74 was dry each year.

Summary and conclusions

These initial efforts indicate that the frequency of drying at a river mile 74 each year can be predicted by the Otowi index supply and, possibly, upstream diversions. The relationship between decreases in river drying and increased diversion amounts may seem counterintuitive. However, it suggests increases in diversions may occur when there is more water in the system and/or agricultural return flows increase with higher diversion rates. Increasing the power to predict the persistence of river mile drying could occur by incorporating corrected evaporation loss estimations, precipitation, agriculture return flows, or other system components not already identified. To predict the occurrence and extent of river drying on the daily time scale will require additional data manipulation, which could include removing seasons for which there are large gaps of missing gage data, parsing the ET Toolbox data into pre and post changes in estimate procedures, and/or applying a correction to the ET Toolbox evaporation estimates. This case study was a small step to understanding what drives river drying and using the Otowi index supply appears to be a good predictor of drying at river mile 74 because it is a reliable and intuitive indicator of the system's available water.

Citations

Akaike, H. (1973). Maximum likelihood identification of Gaussian autoregressive moving average models. *Biometrika*, 60(2), 255–265.

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Srinivasan, V., Thompson, S., Madhyastha, K., Penny, G., Jeremiah, K., & Lele, S. (2015). Why is the Arkavathy River drying? A multiple-hypothesis approach in a data-scarce region. *Hydrology and Earth System Sciences*, 19(4), 1905-1917.

Appendices I-III

Appendix I. – ET Toolbox

The Bureau of Reclamation produces, and curates estimates of water consumption within the Rio Grande Basin (Brower 2018). This is called the Evapotranspiration Toolbox (ET Toolbox) and since 2000 has provided daily high-resolution estimates of rainfall and water depletions (crop and riparian vegetation evapotranspiration, and open water evaporation) for reaches of the river (Figure A1). The data and formula to derive these estimates changed in 2011. Prior to 2011, the Modified-Penman method was used and incorporated minimum and maximum temperature, average relative humidity, average wind speed, and solar radiation for the previous 24-hours. In 2013, the method was changed to the temperature based 1985 Hargreaves-Samani method and all of 2012 was recalculated through the 1985 Hargreaves-Samani method. This methodological change resulted in divergent evaporation estimates, which was not corrected by the addition of simple mean baseline (Figure A2). To produce better fitting models with greater predictive capacity, incorporation of the full and corrected dataset would be important. This could be done by recalculating a representative sample of the old data with the 1985 Hargreaves-Samani method and then derive a relationship between the two, or vice versa.

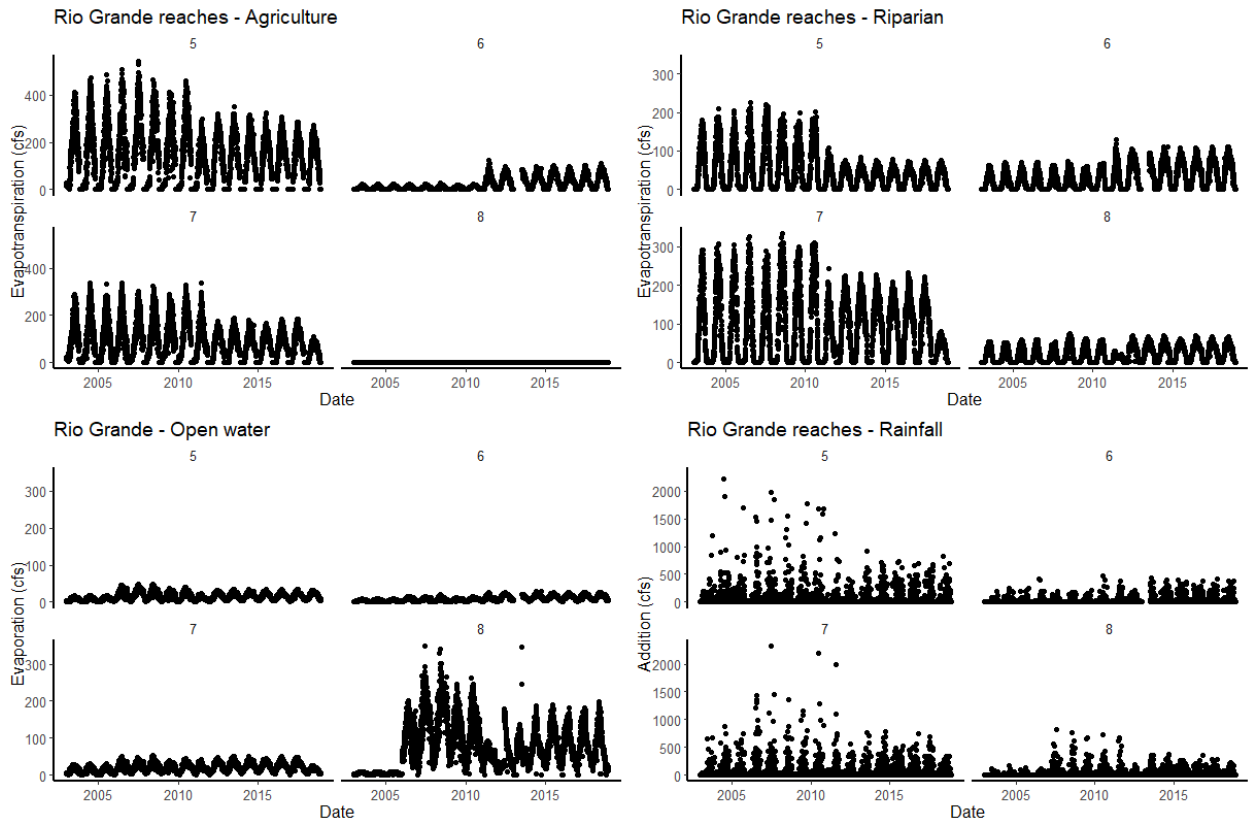

Figure A1. U.S. Bureau of Reclamation's ET Toolbox estimated agriculture, riparian, and open water evaporations, and rainfall additions in Rio Grande reaches 5 to 8 for the years 2003-2018.

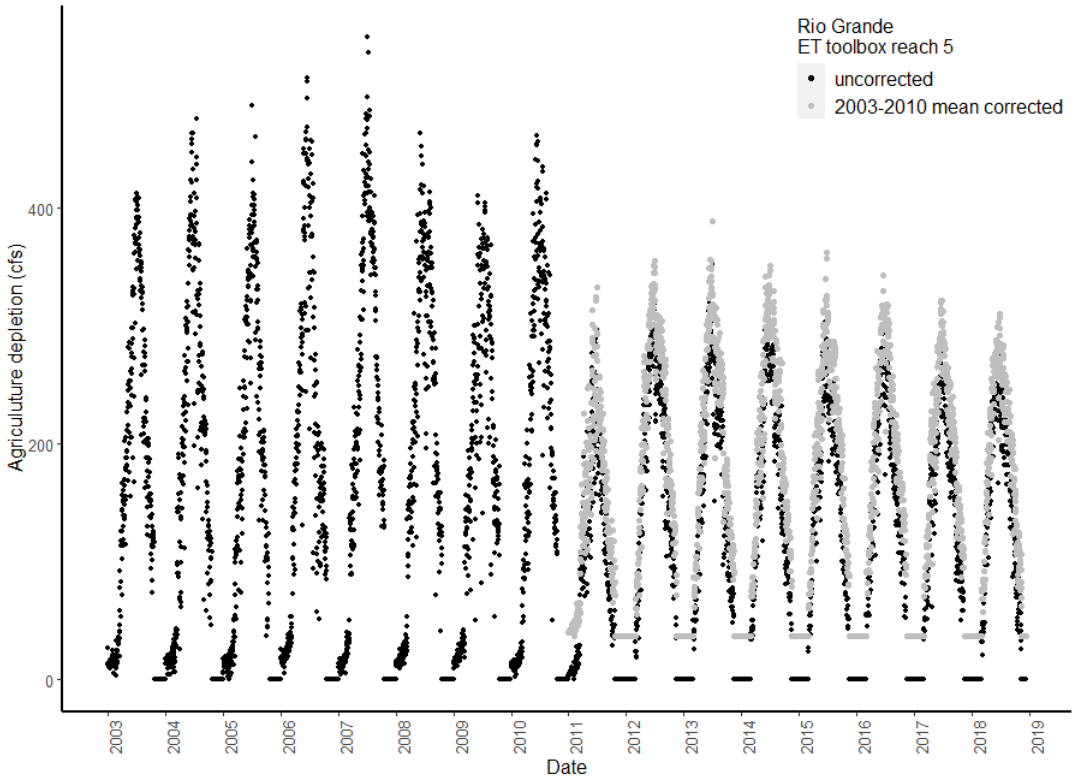

Figure A2. Example of adding a mean baseline correction to the U.S. Bureau of Reclamation's ET Toolbox evapotranspiration from agriculture. Data prior to 2011 was calculated using Modified-Penman and afterwards, the 1985 Hargreaves-Samani method was used.

Appendix II. - Modeling workflow and result to predict the persistence of drying at river mile 74

1) Tested predictors for correlation ('pairs.panels' from and 'psych' package) and retained diversions at the Isleta and San Acacia diversion dams as well as the Otowi index supply (Figure A3)
2) Constructed all model ('lme' function from 'lme4' package) combinations with year as a random effect and assessed model assumptions (Table A1 and Figure A4). Predictor variables were scaled.
3) Calculated AICc and BIC to identify best fitted models (Table A2)
4) Used confidence intervals and analysis of variance to identify significant predictors within each model (Table A3)
5) Reduced best model sets to one that used Isleta diversion dam and one that used the Otowi index supply; estimated effect each predictor using confidence intervals with unscaled data (Table 2)
6) Visualized the effect magnitude for best models (Figure 4)

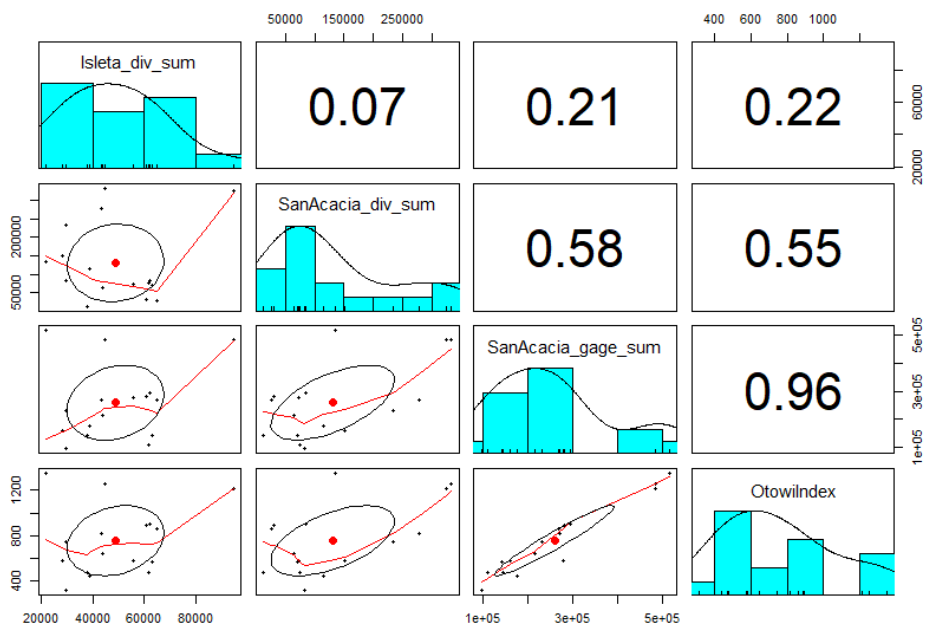

Figure A3. Pairs plot of correlations between predictor variables.

Table A1. Linear mixed-effects (lme) models and assessment of model assumptions used to predict the persistence of drying in Rio Grande at mile 74.

Response	Fixed effects	Random effects	Passed assumptions
River mile 74	Isleta + San Acacia + Otowi	Year	Yes (Figure A2)
	Isleta + San Acacia		
	San Acacia + Isleta		
	San Acacia + Otowi		
	Isleta		
	San Acacia		
	Otowi		

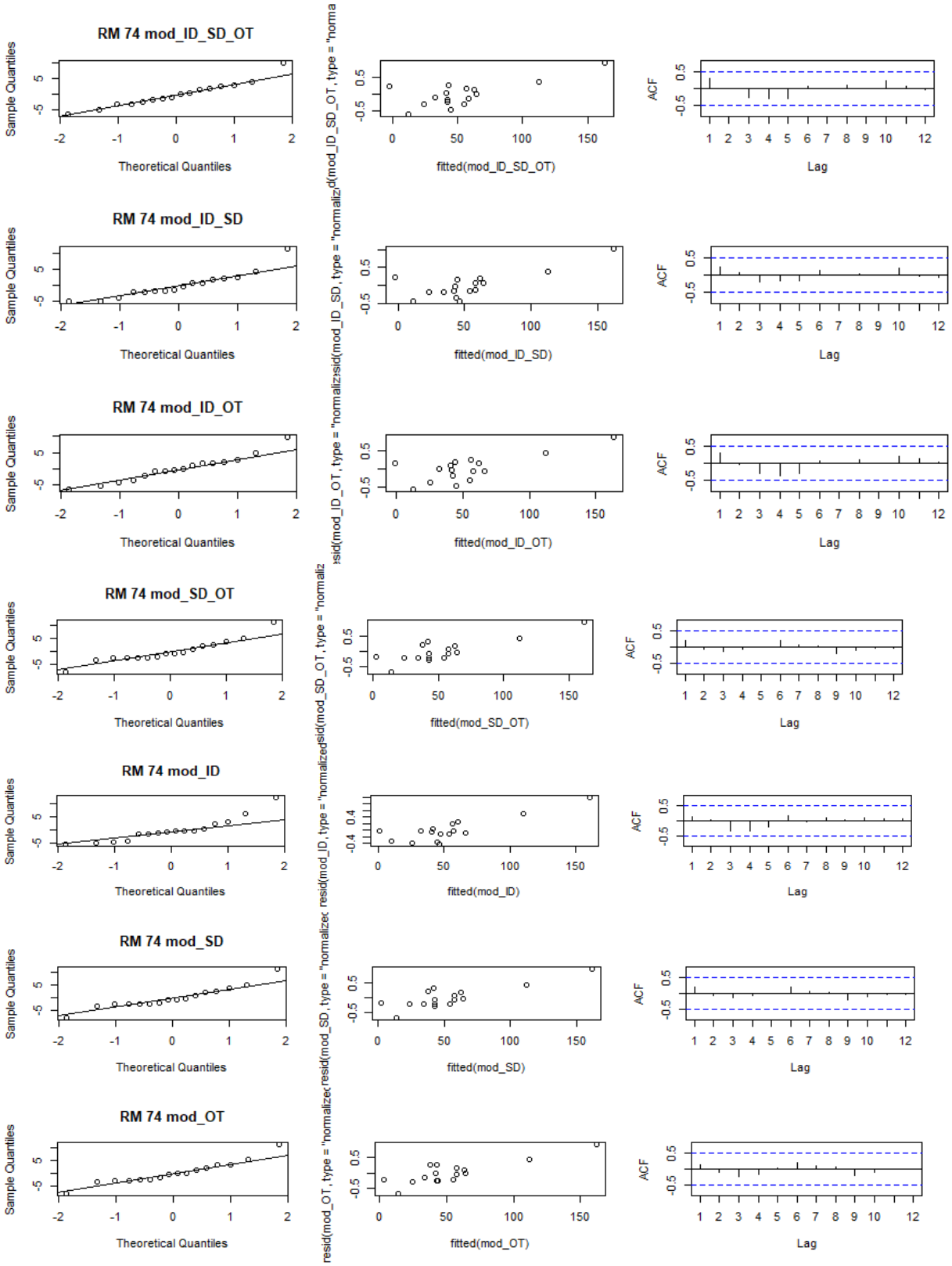

Figure A2. Normality (qqplot and Shapiro-wilks test), heteroscedasticity (fitted versus residuals), and temporal autocorrelation for models predicting the number of days river mile 74 was dry using additive linear mixed-effects models of Isleta diversion (ID), San Acacia diversion (SD) and Otowi supply index (OT) as predictor variables.

Table A3. AICc, BIC, and degrees of freedom (df) results from modeling number of days river mile 74 was dry each year using Isleta diversion, San Acacia diversion, and Otowi index supply as predictor variables. Interpreted models (i.e. a difference of < 2.0 from the top model) highlighted in grey.

Models river mile 74	dAICc	df	Models river mile 74	dBIC	df
Isleta + Otowi	0	4	Isleta + Otowi	0	5
Isleta + San Acacia	0	4	San Acacia	0.9	4
Isleta + San Acacia + Otowi	0.7	5	Otowi	1	4
Isleta	0.9	3	Isleta + San Acacia	1.1	5
Otowi	1.7	5	Isleta + San Acacia + Otowi	2.2	6
San Acacia + Otowi	4.1	5	null	2.7	3
San Acacia	4.1	5	San Acacia + Otowi	3.5	5
null	5.4	6	San Acacia	3.5	5

Table A4. Significance of scaled predictors in models interpreted to predict the persistence of drying at Rio Grande mile 74. Significant predictors identified as estimates whose confidence intervals (CI) do not overlap with zero and supported with analysis of variance p-value (significance identified with an asterisk).

Models river mile 74	Predictor	Lower CI	Upper CI	Estimated effect	ANOVA p-value
Isleta + San Acacia + Otowi	Isleta	-35.1	0.4	-17.3	0.03*
	San Acacia	-27.9	13.6	- 7.1	0.45
	Otowi	-34.0	8.5	-12.8	0.19
Isleta + San Acacia	Isleta	-37.7	-1.8	-19.8	0.02*
	San Acacia	-32.0	3.9	-14.1	0.09
Isleta + Otowi	Isleta	-34.7	0.9	-17.0	0.04*
	Otowi	-24.6	1.0	-17.0	0.04*
Isleta	Isleta	-40.0	-1.3	-20.7	0.02
San Acacia	San Acacia	-36.1	5.4	-15.4	0.11
Otowi	Otowi	-40.0	-1.2	-20.6	0.02*

Appendix III. – Modeling output for individual river miles 155, 134, and 80.

1) Constructed all model ('lme' function from 'lme4' package) combinations with year as a random effect and assessed model assumptions for individual river miles 155, 134, and 80.
    a) Assumptions for river miles 155 and 80 were met with through qqplot visualization and Shapiro-wilk's test of ~0.9, heteroscedasticity was relatively random, and temporal autocorrelation absent (Figures A5 and A7)
    b) Assumption for river mile 134 were not substantiated by qqplot visualization and Shapiro-wilk's tests ranged from 0.51-0.62. Heteroscedasticity appeared clumped but was driven by a single data point. The assumption of no temporal autocorrelation was supported (Figure A6)
2) Calculated AICc and BIC for river miles 155 and used confidence intervals and analysis of variance to identify models with significant predictors (Tables A5 and A6).
3) Calculated AICc and BIC for river miles 80 and used confidence intervals and analysis of variance to identify models with significant predictors (Tables A7 and A8).

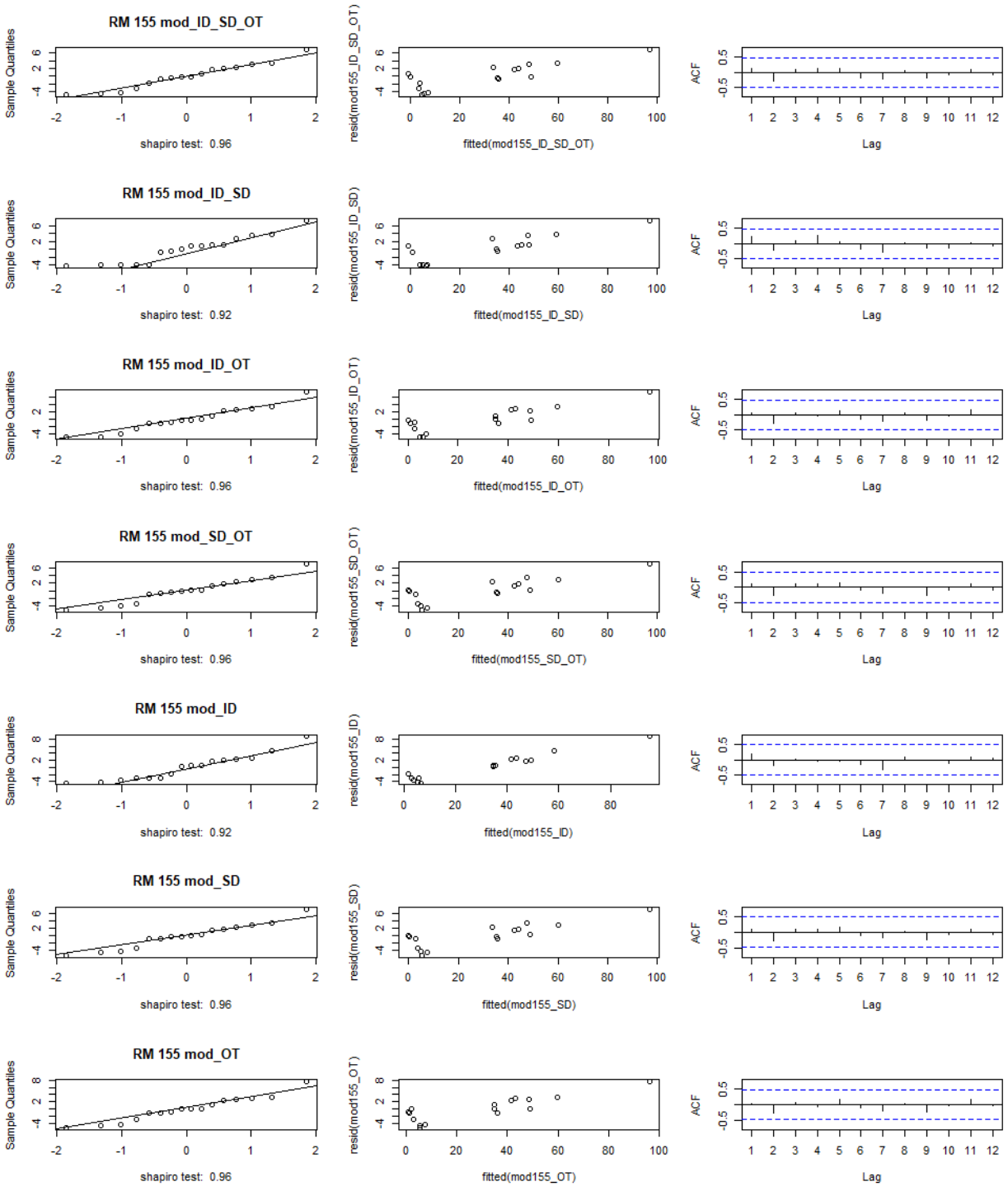

Figure A5. Normality (Shapiro test), heteroscedasticity, and temporal autocorrelation for models predicting number of dry river mile 155 days using additive linear mixed-effects Isleta diversion (ID), San Acacia diversion (SD) and Otowi supply index (OT) as predictor variables.

Table A5. AICc, BIC, and degrees of freedom (df) results from modeling annual number of days river mile 155 was dry using Isleta diversion, San Acacia diversion, and Otowi supply index as predictor variables. Interpreted models highlighted in grey.

Models river mile 155	dAICc	Df	Model river mile 155	dBIC	df
Otowi	0	4	Otowi	0	4
Null	0.2	3	null	1.1	3
Isleta	3.2	4	San Acacia + Otowi	2	5
San Acacia + Otowi	3.6	5	San Acacia	2	5
San Acacia	3.6	5	Isleta + Otowi	2.6	5
Isleta + Otowi	4.2	5	Isleta + San Acacia	3	5
Isleta + San Acacia	4.6	5	Isleta	3.2	4
Isleta + San Acacia + Otowi	8.7	6	Isleta + San Acacia + Otowi	4.5	6

Table A6. Significance of scaled predictors in models interpreted to forecast persistence of drying at Rio Grande mile 155. Significant prediction identified as estimates whose confidence intervals (CI) did not overlap was not present. Analysis of variance p-value was significant and identified with an asterisk.

Models river mile 155	Predictor	Lower CI	Upper CI	Estimated effect	ANOVA p-value
Otowi	Otowi	-28.5	0.37	-14.1	0.04*

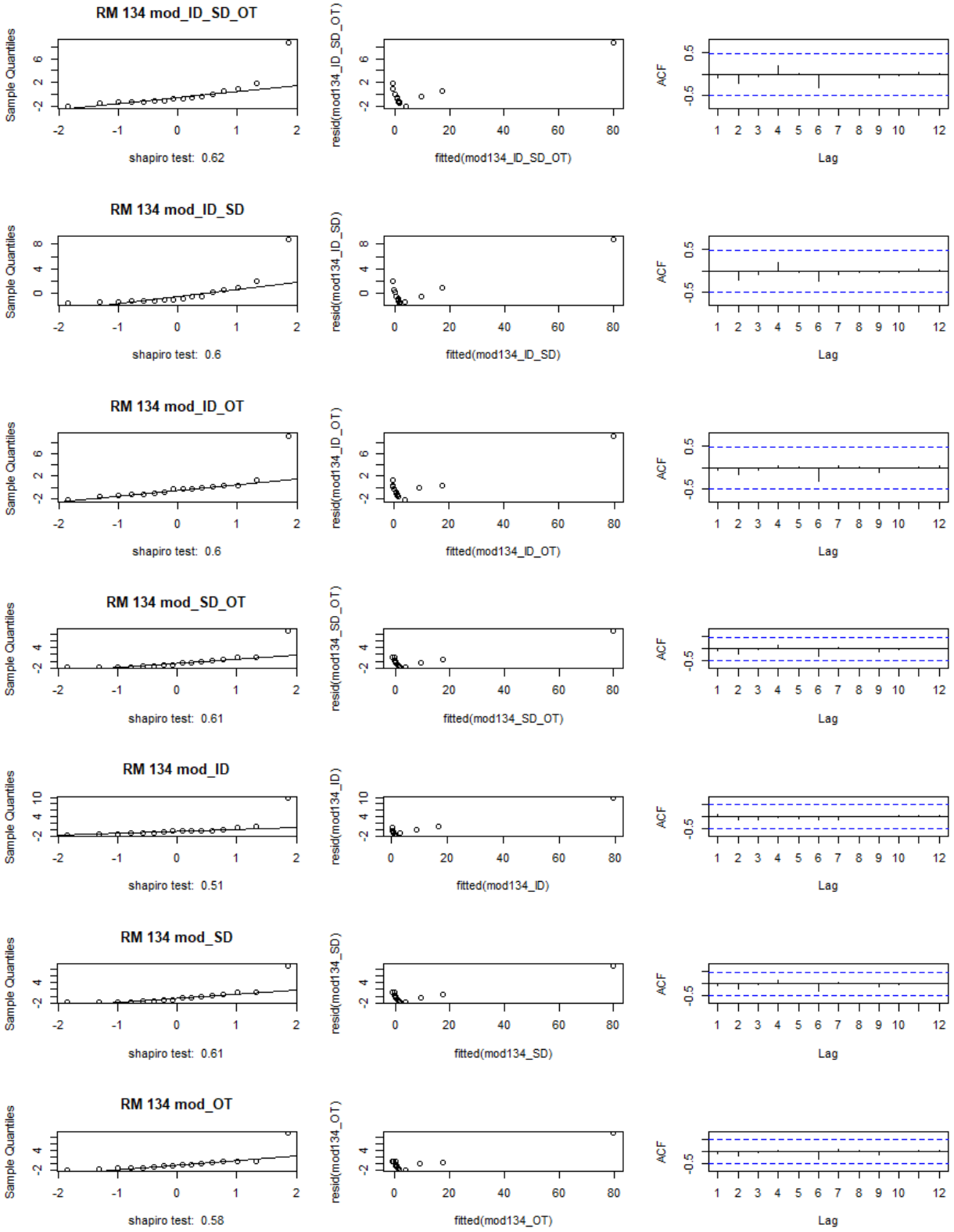

Figure A6. Normality (Shapiro test), heteroscedasticity, and temporal autocorrelation for models predicting number of dry river mile 134 days using additive linear mixed-effects Isleta diversion (ID), San Acacia diversion (SD) and Otowi supply index (OT) as predictor variables.

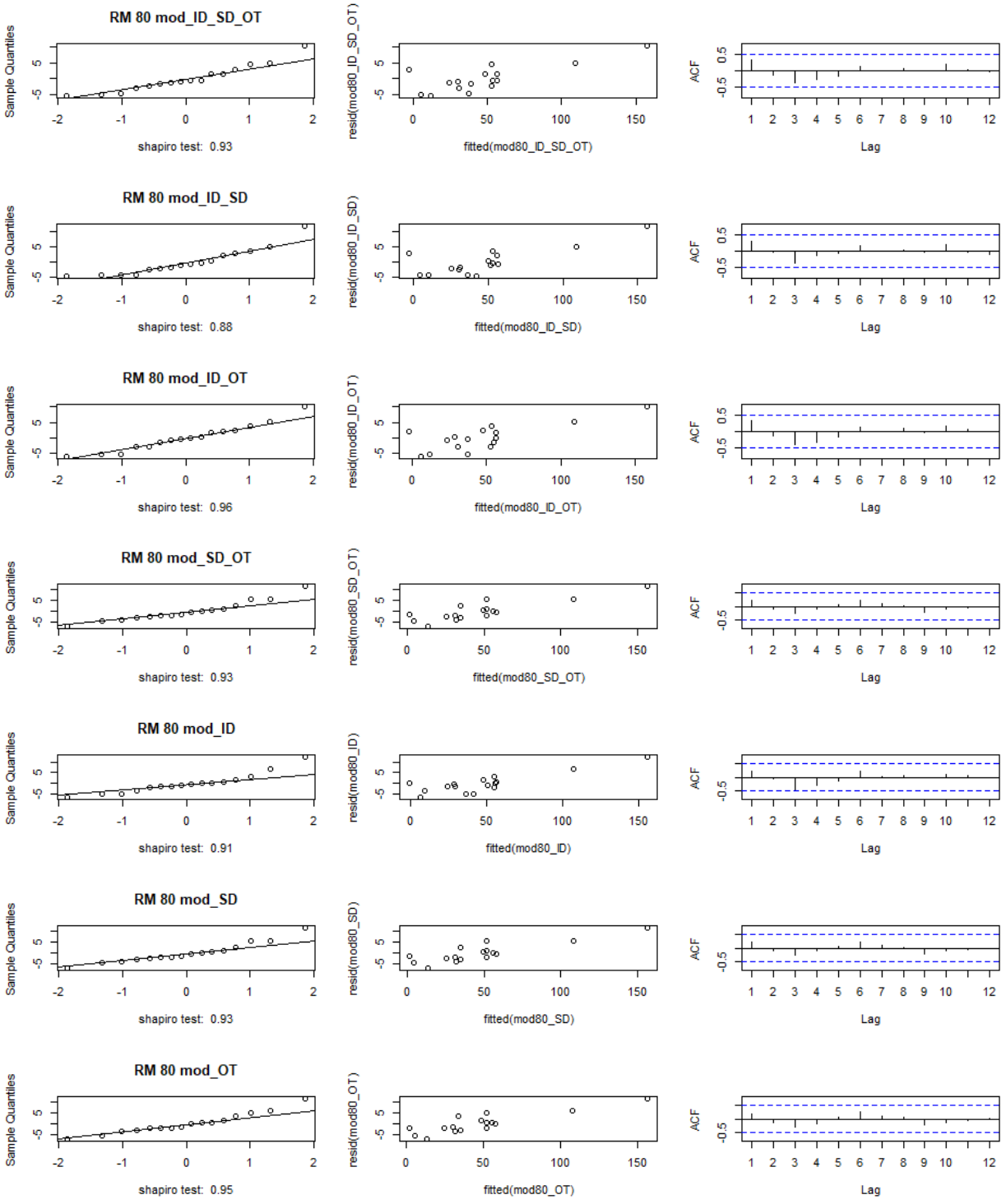

Figure A7. Normality (Shapiro test), heteroscedasticity, and temporal autocorrelation for models predicting number of dry river mile 80 days using additive linear mixed-effects Isleta diversion (ID), San Acacia diversion (SD) and Otowi supply index (OT) as predictor variables.

Table A7. AICc, BIC, and degrees of freedom (df) results from modeling annual number of days river mile 80 was dry using Isleta diversion, San Acacia diversion, and Otowi supply index as predictor variables. Interpreted models highlighted in grey.

Models river mile 80	dAICc	Df	Model river mile 80	dBIC	df
Isleta	0	4	Isleta + Otowi	0	5
Otowi	0.4	4	Isleta	0.1	4
null	0.4	3	Otowi	0.5	4
Isleta + Otowi	1.5	5	Isleta + San Acacia	0.8	5
Isleta + San Acacia	2.3	5	null	1.4	3
San Acacia + Otowi	4.5	5	Isleta + San Acacia + Otowi	2.3	6
San Acacia	4.5	5	San Acacia + Otowi	3	5
Isleta + San Acacia + Otowi	6.4	6	San Acacia	3	5

Table A8. Significance of scaled predictors in models interpreted to forecast persistence of drying at Rio Grande mile 80. Significant prediction identified as estimates whose confidence intervals (CI) did not overlap with zero highlighted in grey and analysis of variance p-value significance identified with an asterisk.

Models river mile 74	Predictor	Lower CI	Upper CI	Estimated effect	ANOVA p-value
Isleta + Otowi	Isleta	-35.7	2.3	-16.7	0.06
	Otowi	-34.5	3.9	-15.5	0.08
Isleta + San Acacia	Isleta	-28.3	-0.29	-19.3	0.03*
	San Acacia	-32.1	5.9	-13.1	0.14
Isleta	Isleta	-40.3	-0.07	-20.2	0.03*
Otowi	Otowi	-34.5	3.4	-15.5	0.04*